No
Date

Nabila Alia Zahra
20134402S

Dik : swr source = 2 : 1
swr detector : 1,2 : 1

Ditanya : Uncertainty dan diagram!

Jawab :

$$\rho_s = \frac{|SWR_s|-1}{|SWR_s|+1}$$

$$\rho_s = \frac{2-1}{2+1} = 0{,}333$$

$$\rho_d = \frac{|SWR_d|-1}{|SWR_d|+1}$$

$$= \frac{1{,}2-1}{1{,}2+1} = 0{,}091$$

a) Power Available

$$(1-\rho_d^2)\,(-)\,\begin{matrix}(-)\\(+)\end{matrix}\,|1 \pm \rho_s\rho_d|^2$$

$$1-\rho_d^2 = 1-(0{,}091)^2$$

$$1-\rho_d^2 = 0{,}992$$

dalam dBm $= 10\log(1-\rho_d^2)$

$1-\rho_d^2 = -0{,}035$ dBm

$$(1+\rho_s\rho_d)^2 = (1+0{,}333\times 0{,}091)^2$$

$$= 1{,}062 = 0{,}261 \text{ dBm}$$

$$(1-\rho_s\rho_d)^2 = (1-0{,}333\times 0{,}091)^2$$

$$= 0{,}94 = -0{,}269 \text{ dBm}$$

$$\Delta \text{ power} = \frac{(1-\rho_d^2)}{(1 \pm \rho_s\rho_d)^2}$$

JOYKO 36 Lines, 6 mm

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$$\Delta P_1 = \frac{(1-\rho d^2)}{(1+\rho s \rho d)^2} = \frac{1-0.091^2}{(1+0.333 \times 0.091)^2}$$

$$= 0.934$$

$$\Delta P_0 = \frac{(1-\rho d^2)}{(1-\rho s \rho d)^2} = \frac{1-0.091^2}{(1-0.333 \times 0.091)} = 1.055$$

$\therefore$ 10 log $\Delta P_1 = -0.1295$ dBm

$\therefore$ 10 log $\Delta P_0 = 0.1231$ dBm

diagram power

